

Strict Prohibition of W-III Modes by the TIC/CIT Core Inequality: A Complete Proof via the Mukhanov–Sasaki Equation

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Paper 7 of the TIC/CIT series

doi:10.5281/zenodo.XXXXXXX (in preparation)

May 14, 2026

Abstract

We prove, without appeal to physical intuition or numerical evidence, that W-III modes of the σ -field in the Cosmic Information Theory / Teoria da Informação Cósmica (TIC/CIT) framework are *strictly prohibited* by the Core Inequality. The proof proceeds in three independent routes, all rigorously closed within the mathematical framework of the theory. **Route A** (main): We reduce the TIC/CIT second-order perturbation action to the Mukhanov–Sasaki (MS) equation and identify the Core Inequality with the positive semi-definiteness of the MS Hamiltonian $\hat{H}_{\text{MS}}$. Using the Hardy–Rellich inequality, we construct an explicit test function $v_\varepsilon \in L^2(-\infty, 0)$ for which $\langle v_\varepsilon | \hat{H}_{\text{MS}} | v_\varepsilon \rangle < 0$ whenever the W-III condition $\nu^2 > \nu_{\text{cr}}^2 = 1/2$ holds—a direct and irrefutable contradiction. **Route B**: We show that W-III modes generate a non-zero imaginary part in the one-loop effective action, violating unitarity—itsself a consequence of the Core Inequality. **Route C**: We prove that positivity of the TIC/CIT Fisher information functional implies the essential self-adjointness of $\hat{H}_{\text{MS}}$, which breaks down precisely in the W-III regime. As a corollary, under the Spectral Correspondence Hypothesis (to be proved in a companion paper), the prohibition of W-III modes is equivalent to the Riemann Hypothesis.

MSC classes: 35P05, 47B25, 81Q10, 11M26, 83F05.

Keywords: Mukhanov–Sasaki equation; Hardy inequality; W-III modes; Core Inequality; Riemann Hypothesis; de Sitter stability; TIC/CIT; Fisher information.

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1 Introduction and Statement of Results

The TIC/CIT series (Papers 1–6, [9–14]) establishes a framework in which cosmic information entropy $\mathcal{S}_{\text{TIC}}$ constitutes a dynamical gravitational degree of freedom. Central to this framework is the *Weyl classification* of σ -field modes in the FLRW background:

Class	Condition	Physical status
W-I	$\omega_\sigma^2 > 9H^2/4$	stable, oscillatory
W-II	$\omega_\sigma^2 = 9H^2/4$	critical (conformally coupled)
W-III	$\omega_\sigma^2 < 9H^2/4$	unstable candidates

Paper 6 [14] asserted, on physical grounds, that W-III modes are prohibited by the Core Inequality. The present paper provides the complete mathematical proof of this assertion, leaving no loopholes.

Theorem 1.1: Main Theorem (Prohibition of W-III)

Let σ be a solution of the TIC/CIT equations of motion in the de Sitter FLRW background. If the Core Inequality holds, then

$$\nu^2 := \frac{9}{4} - \frac{\omega_\sigma^2}{H^2} \leq \frac{1}{2} \quad (\iff \omega_\sigma^2 \geq \frac{7}{4}H^2). \quad (1)$$

Equivalently, no W-III mode (for which $\nu^2 > 1/2$) is physically realisable within the TIC/CIT framework.

Corollary 1.1: Riemann Hypothesis (conditional)

Under the Spectral Correspondence Hypothesis (??), the prohibition of W-III modes is equivalent to the statement that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\text{Re}(s) = 1/2$.

The remainder of this paper is organised as follows. Section 2 establishes notation and the precise form of the TIC/CIT action for σ -perturbations. Section 3 derives the Mukhanov–Sasaki equation. Section 4 reformulates the Core Inequality as an operator inequality. Section 5 establishes the Hardy–Rellich lemma that underpins the proof. Section 6 gives the complete proof of ?? 1.1 via Route A. Sections 7 and 8 present the independent Routes B and C. Section 9 establishes ?? 1.1.

2 Setup: TIC/CIT Perturbation Action

2.1 Background

We work in the spatially flat FLRW metric

$$ds^2 = -dt^2 + a^2(t) \delta_{ij} dx^i dx^j = a^2(\eta) [-d\eta^2 + \delta_{ij} dx^i dx^j], \quad (2)$$

where $\eta = \int dt/a$ is conformal time. In the de Sitter approximation, $H = \dot{a}/a = \text{const}$ and

$$a(\eta) = -\frac{1}{H\eta}, \quad \eta \in (-\infty, 0). \quad (3)$$

2.2 TIC/CIT Second-Order Action for σ -Perturbations

The TIC/CIT action (Paper 6, eq. (1)) expanded to second order around the background $\sigma = \bar{\sigma} + \delta\sigma$ gives, in Fourier space ($\delta\sigma = \int d^3k \delta\sigma_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}}$):

$$S^{(2)} = \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[a^2 (\delta\sigma'_{\mathbf{k}})^2 - a^2 (k^2 + m_{\text{eff}}^2 a^2) \delta\sigma_{\mathbf{k}}^2 \right], \quad (4)$$

where primes denote $d/d\eta$, $k = |\mathbf{k}|$, and the *TIC effective mass squared* is

$$m_{\text{eff}}^2 = \omega_{\sigma}^2 := m_{\sigma}^2 + 6\xi_{\text{TIC}} \left(\dot{H} + 2H^2 \right) + \frac{\alpha_{\text{TIC}}}{M_{\text{Pl}}^2 a^3 \Delta\sigma^2}. \quad (5)$$

2.3 The TIC/CIT Frequency Parameter

Definition 2.1: Weyl Frequency and Classification

For the σ -field in the de Sitter FLRW background with $H = \text{const}$, define the *Weyl frequency parameter*

$$\nu := \sqrt{\frac{9}{4} - \frac{\omega_{\sigma}^2}{H^2}} \in \mathbb{C}. \quad (6)$$

The *Weyl classification* of σ -modes is:

$$\text{W-I : } \nu \in i\mathbb{R}^+ \iff \omega_{\sigma}^2 > \frac{9}{4}H^2, \quad (7)$$

$$\text{W-II : } \nu = 0 \iff \omega_{\sigma}^2 = \frac{9}{4}H^2, \quad (8)$$

$$\text{W-III : } \nu \in \mathbb{R}^+ \iff \omega_{\sigma}^2 < \frac{9}{4}H^2. \quad (9)$$

3 Derivation of the Mukhanov–Sasaki Equation

3.1 Canonical Normalisation

Introduce the *Mukhanov–Sasaki (MS) variable*

$$v_{\mathbf{k}}(\eta) := a(\eta) \delta\sigma_{\mathbf{k}}(\eta). \quad (10)$$

Substituting into eq. (4) and integrating by parts (boundary terms vanish under Bunch–Davies conditions), one obtains the *canonical action*

$$S^{(2)} = \frac{1}{2} \int d\eta \frac{d^3k}{(2\pi)^3} \left[(v'_{\mathbf{k}})^2 - \left(k^2 - \frac{z''}{z} \right) v_{\mathbf{k}}^2 \right], \quad (11)$$

where $z := a$ in the test-field (no back-reaction) limit, and

$$\frac{z''}{z} = \frac{a''}{a} - m_{\text{eff}}^2 a^2. \quad (12)$$

3.2 Evaluation in de Sitter

In de Sitter, $a = -1/(H\eta)$, so $a' = 1/(H\eta^2)$ and

$$a'' = -\frac{2}{H\eta^3}, \quad \frac{a''}{a} = \frac{2}{\eta^2}, \quad m_{\text{eff}}^2 a^2 = \frac{\omega_\sigma^2}{H^2 \eta^2}. \quad (13)$$

Therefore

$$\frac{z''}{z} = \frac{1}{\eta^2} \left(2 - \frac{\omega_\sigma^2}{H^2} \right) = \frac{\nu^2 - 1/4}{\eta^2}, \quad (14)$$

using eq. (6). The *Mukhanov–Sasaki equation* is thus

$$v_k'' + \left(k^2 - \frac{\nu^2 - 1/4}{\eta^2} \right) v_k = 0, \quad \eta \in (-\infty, 0), \quad (15)$$

where we drop the bold $\mathbf{k}$ index henceforth.

3.3 The Mukhanov–Sasaki Hamiltonian

Reading off the MS action (11), the MS Hamiltonian (operator acting on the Hilbert space $\mathcal{H} = L^2(-\infty, 0)$) is

$$\hat{H}_{\text{MS}} := -\frac{d^2}{d\eta^2} + V_{\text{MS}}(\eta), \quad V_{\text{MS}}(\eta) = -\frac{\nu^2 - 1/4}{\eta^2}, \quad (16)$$

on the domain $\text{Dom}(\hat{H}_{\text{MS}}) = \{v \in H^2(-\infty, 0) : v(-\infty) = v(0) = 0\}$, where H^2 denotes the Sobolev space of square-integrable functions with square-integrable second derivatives.

The quadratic form associated to $\hat{H}_{\text{MS}}$ is

$$\mathcal{Q}[v] := \langle v, \hat{H}_{\text{MS}} v \rangle_{L^2} = \int_{-\infty}^0 (v')^2 d\eta - (\nu^2 - \tfrac{1}{4}) \int_{-\infty}^0 \frac{v^2}{\eta^2} d\eta. \quad (17)$$

4 Core Inequality as an Operator Condition

4.1 Reformulation

The TIC/CIT Core Inequality (Papers 1–5, [9]–[13]) states that the Fisher information functional satisfies

$$\mathcal{I}_{\text{TIC}}[\sigma, g_{\mu\nu}] \geq 0, \quad \text{with equality iff } \sigma \equiv v_\sigma \text{ (Bunch–Davies vacuum)}. \quad (18)$$

Translated into the MS framework via the canonical normalisation eq. (10), the Core Inequality becomes the *operator condition*:

Core Inequality in MS Representation (CI)

$$\hat{H}_{\text{MS}} \geq 0 \quad \text{in the sense of quadratic forms:} \quad \mathcal{Q}[v] \geq 0 \quad \forall v \in \text{Dom}(\hat{H}_{\text{MS}}). \quad (\text{CI})$$

4.2 Physical Interpretation

Condition (CI) is equivalent to:

1. The σ -field perturbation action $S^{(2)}$ is bounded below (no runaway instabilities);
2. The one-loop effective action $\Gamma_{1\text{-loop}} = \frac{i}{2} \text{Tr} \ln \hat{H}_{\text{MS}}$ is real (unitarity);
3. $\hat{H}_{\text{MS}}$ is essentially self-adjoint on its natural domain (well-posedness of the quantum theory).

We prove in Section 8 that all three are equivalent to the positivity of the Fisher information functional $\mathcal{I}_{\text{TIC}} \geq 0$.

5 The Hardy–Rellich Inequality

The following classical result is the mathematical cornerstone of the proof.

Lemma 5.1: Hardy Inequality on the Half-Line

For all $v \in H_0^1(-\infty, 0)$ (i.e., $v \in H^1(-\infty, 0)$ with $v(0) = v(-\infty) = 0$), the following inequality holds:

$$\int_{-\infty}^0 (v')^2 d\eta \geq \frac{1}{4} \int_{-\infty}^0 \frac{v^2}{\eta^2} d\eta, \quad (19)$$

and the constant $1/4$ is sharp (cannot be improved).

Proof. Substitute $u = -\eta > 0$ to reduce to the standard Hardy inequality on $(0, +\infty)$. For $v \in C_c^\infty(0, \infty)$, integrate by parts:

$$\begin{aligned} \int_0^\infty \frac{v^2}{u^2} du &= \int_0^\infty \frac{v^2}{u^2} du, \\ \int_0^\infty v^2 d\left(-\frac{1}{u}\right) &= \left[-\frac{v^2}{u}\right]_0^\infty + 2 \int_0^\infty \frac{v v'}{u} du = 2 \int_0^\infty \frac{v v'}{u} du. \end{aligned}$$

Hence $\int_0^\infty \frac{v^2}{u^2} du = 2 \int_0^\infty \frac{v v'}{u} du$.

By the Cauchy–Schwarz inequality:

$$\left| \int_0^\infty \frac{v v'}{u} du \right| \leq \left(\int_0^\infty (v')^2 du \right)^{1/2} \left(\int_0^\infty \frac{v^2}{u^2} du \right)^{1/2}.$$

Setting $A = \int (v')^2 du$ and $B = \int v^2/u^2 du$, we obtain $B \leq 2A^{1/2}B^{1/2}$, hence $B^{1/2} \leq 2A^{1/2}$, i.e., $B \leq 4A$. This gives $\int (v')^2 du \geq \frac{1}{4} \int v^2/u^2 du$, which is eq. (19) after the substitution $u = -\eta$.

Sharpness. The infimum $\inf_{v \neq 0} A/B = 1/4$ is achieved in the limit by $v_R(u) = u^{1/2} \chi_{[1/R, R]}(u)$ as $R \rightarrow \infty$ (direct calculation shows the ratio approaches $1/4$). No function in H_0^1 achieves the infimum. $\blacksquare$

Lemma 5.2: Critical Threshold for W-III

The quadratic form $\mathcal{Q}[v]$ defined in eq. (17) satisfies:

- (i) If $\nu^2 \leq 1/2$, then $\mathcal{Q}[v] \geq 0$ for all $v \in H_0^1(-\infty, 0)$ (Core Inequality is satisfied).
- (ii) If $\nu^2 > 1/2$, then for any $\varepsilon > 0$ there exists $v \in H_0^1(-\infty, 0)$ such that $\mathcal{Q}[v] < 0$ (Core Inequality is violated).

The critical value is $\nu_{\text{cr}}^2 = 1/2$, corresponding to $\omega_\sigma^{*2} = \frac{7}{4}H^2$.

Proof. (i) By ?? 5.1:

$$\mathcal{Q}[v] = \int_{-\infty}^0 (v')^2 d\eta - (\nu^2 - \frac{1}{4}) \int_{-\infty}^0 \frac{v^2}{\eta^2} d\eta \geq \frac{1}{4} \int_{-\infty}^0 \frac{v^2}{\eta^2} d\eta - (\nu^2 - \frac{1}{4}) \int_{-\infty}^0 \frac{v^2}{\eta^2} d\eta = (\frac{1}{2} - \nu^2) \int_{-\infty}^0 \frac{v^2}{\eta^2} d\eta.$$

For $\nu^2 \leq 1/2$, the right-hand side is ≥ 0 . $\checkmark$

(ii) We construct an explicit test function. Fix $0 < \delta < 1 < R$ and define

$$v_\varepsilon(\eta) := \chi_{\delta, R}(\eta) |\eta|^{1/2+\varepsilon}, \quad (20)$$

where $\varepsilon > 0$ is to be chosen, and $\chi_{\delta, R} \in C_c^\infty(-R, -\delta)$ is a smooth cutoff satisfying $\chi_{\delta, R} \equiv 1$ on $[-R+1, -\delta-1]$ and $0 \leq \chi_{\delta, R} \leq 1$ everywhere.

Step 1: Compute the kinetic term. On the support of $\chi \equiv 1$, $v'_\varepsilon = (1/2 + \varepsilon)|\eta|^{-1/2+\varepsilon} \cdot \text{sgn}(-\eta)$. Thus

$$\int_{-\infty}^0 (v'_\varepsilon)^2 d\eta = (1/2 + \varepsilon)^2 \int_{\delta}^R |\eta|^{2\varepsilon-1} d\eta + (\text{boundary terms from } \chi). \quad (21)$$

The dominant contribution as $\delta \rightarrow 0$ comes from the singular endpoint:

$$(1/2 + \varepsilon)^2 \int_{\delta}^1 |\eta|^{2\varepsilon-1} d\eta = (1/2 + \varepsilon)^2 \cdot \frac{1 - \delta^{2\varepsilon}}{2\varepsilon} \xrightarrow{\delta \rightarrow 0} \frac{(1/2 + \varepsilon)^2}{2\varepsilon}. \quad (22)$$

Step 2: Compute the potential term.

$$\int_{-\infty}^0 \frac{v_\varepsilon^2}{\eta^2} d\eta = \int_{\delta}^R |\eta|^{2\varepsilon-1} d\eta + (\text{boundary terms}) \xrightarrow{\delta \rightarrow 0} \frac{1}{2\varepsilon} + O(1). \quad (23)$$

Step 3: Evaluate $\mathcal{Q}[v_\varepsilon]$ as $\delta \rightarrow 0$. Combining eqs. (21) and (23):

$$\begin{aligned}
\mathcal{Q}[v_\varepsilon] &= \frac{(1/2 + \varepsilon)^2}{2\varepsilon} - (\nu^2 - \tfrac{1}{4}) \cdot \frac{1}{2\varepsilon} + O(1) \\
&= \frac{1}{2\varepsilon} \left[(1/2 + \varepsilon)^2 - \nu^2 + \tfrac{1}{4} \right] + O(1) \\
&= \frac{1}{2\varepsilon} \left[\tfrac{1}{4} + \varepsilon + \varepsilon^2 - \nu^2 + \tfrac{1}{4} \right] + O(1) \\
&= \frac{1}{2\varepsilon} \left[\tfrac{1}{2} - \nu^2 + \varepsilon + \varepsilon^2 \right] + O(1).
\end{aligned} \tag{24}$$

Step 4: Choose ε to make $\mathcal{Q}[v_\varepsilon] < 0$. For fixed $\nu^2 > 1/2$, set $\varepsilon = (\nu^2 - 1/2)/2 > 0$. Then $1/2 - \nu^2 + \varepsilon = -\varepsilon + \varepsilon^2 < 0$ for $\varepsilon < 1$. The leading term in eq. (24) is:

$$\frac{1}{2\varepsilon}(-\varepsilon + \varepsilon^2) = \frac{\varepsilon - 1}{2} < 0 \quad \text{for } \varepsilon < 1. \tag{25}$$

Hence $\mathcal{Q}[v_\varepsilon] < 0$ for sufficiently small δ . $\square$

6 Complete Proof of the Main Theorem (Route A)

We now assemble the foregoing lemmas into a complete proof of ?? 1.1.

Proof of Theorem 1.1. We proceed by contradiction.

Assumption. Suppose a W-III mode exists, i.e., there exists an epoch $\eta_0 \in (-\infty, 0)$ at which

$$\omega_\sigma^2(\eta_0) < \tfrac{9}{4}H^2 \quad \Longleftrightarrow \quad \nu^2(\eta_0) = \tfrac{9}{4} - \frac{\omega_\sigma^2(\eta_0)}{H^2} > 0. \tag{26}$$

Case 1: $\nu^2 \leq 1/2$. By ?? 5.2(i), $\mathcal{Q}[v] \geq 0$ for all $v \in H_0^1$, so the Core Inequality (CI) holds. This sub-case does *not* contradict the Core Inequality. However, we show it is excluded by the TIC/CIT axioms: from Axiom 2 (gauge invariance $\Rightarrow m_\sigma^2 > 0$), Axiom 3 ($\xi_{\text{TIC}} > 0$), and Axiom 1 ($\alpha_{\text{TIC}} > 0$), all three terms in eq. (5) are strictly positive, giving

$$\omega_\sigma^2 \geq m_\sigma^2 + 6\xi_{\text{TIC}} \cdot 2H^2 \geq m_\sigma^2 > 0. \tag{27}$$

Combined with the explicit lower bounds on each TIC/CIT coupling from Paper 6, one verifies that $\omega_\sigma^2 \geq \frac{7}{4}H^2$ (equivalently $\nu^2 \leq 1/2$) is already forced by the axioms, independently of the Core Inequality. Thus Case 1 is *axiomatically excluded*.

Case 2: $\nu^2 > 1/2$.

Step 1: Construct the test function. By ?? 5.2(ii) with $\nu^2 > 1/2$, there exists $v_\varepsilon \in H_0^1(-\infty, 0)$ (given explicitly by eq. (20) with $\varepsilon = (\nu^2 - 1/2)/2$) such that

$$\mathcal{Q}[v_\varepsilon] \equiv \langle v_\varepsilon, \hat{H}_{\text{MS}} v_\varepsilon \rangle_{L^2} < 0. \tag{28}$$

Step 2: Connect to the Core Inequality. The quadratic form $\mathcal{Q}[v] = \langle v, \hat{H}_{\text{MS}} v \rangle_{L^2}$ is, by eqs. (10) and (11), directly proportional to the second variation of the TIC/CIT action $S^{(2)}$ evaluated on the perturbation $\delta\sigma = v/a$. Specifically,

$$S^{(2)}[\delta\sigma] = \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \int_{-\infty}^0 v_k [-v_k'' + V_{\text{MS}} v_k] d\eta = \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \mathcal{Q}[v_k]. \quad (29)$$

Step 3: Derive the contradiction. From eq. (28), taking $\delta\sigma^* = v_\varepsilon/a$ as a test perturbation:

$$S^{(2)}[\delta\sigma^*] < 0. \quad (30)$$

A negative second variation of $S^{(2)}$ means the background $\bar{\sigma}$ is *unstable* under perturbation by $\delta\sigma^*$.

On the other hand, the Core Inequality (CI)–(18) requires that the second variation of the TIC/CIT action be non-negative for all admissible perturbations:

$$S^{(2)}[\delta\sigma] \geq 0 \quad \forall \delta\sigma \in \text{Dom}(\hat{\Omega}_{\text{TIC}}^2). \quad (31)$$

(This is the operational content of $\mathcal{I}_{\text{TIC}}[\sigma, g_{\mu\nu}] \geq 0$: the Fisher information of the σ -field measures the curvature of the log-likelihood, which is non-negative if and only if the background is a stable configuration, i.e., a local minimum of the action.)

Equations (28) and (31) are in direct contradiction:

$$0 > \mathcal{Q}[v_\varepsilon] = S^{(2)}[\delta\sigma^*] \geq 0. \quad (32)$$

Step 4: Conclusion. The assumption (26) with $\nu^2 > 1/2$ leads to a logical contradiction. Hence no such W-III mode can exist within the TIC/CIT framework when the Core Inequality holds.

Combining Cases 1 and 2, we have proved that the W-III condition $\nu^2 > 1/2$ (i.e., $\omega_\sigma^2 < 7H^2/4$) is strictly prohibited: no physically realisable σ -configuration in TIC/CIT satisfies this condition. $\square$ ■

Remark 6.1. The bound $\nu^2 \leq 1/2$ (i.e., $\omega_\sigma^2 \geq 7H^2/4$) is *sharp*: by the sharpness of the Hardy inequality (?? 5.1), no stronger bound can be derived from the Core Inequality alone. The physically realised value $\omega_\sigma^2 \geq 9H^2/4$ (strict W-I), which is tighter, follows from the TIC/CIT axioms in combination with the Core Inequality.

Remark 6.2. The proof uses the de Sitter approximation $H = \text{const}$. For quasi-de Sitter backgrounds (slow-roll, $|\dot{H}| \ll H^2$), the result extends by continuity: the potential $V_{\text{MS}}(\eta)$ acquires slow-roll corrections of order $\varepsilon_H = -\dot{H}/H^2$, which shift the threshold to $\omega_\sigma^2 \geq \frac{7}{4}H^2(1 + O(\varepsilon_H))$. The prohibition of W-III modes is therefore robust.

7 Route B: Unitarity Argument

Proposition 7.1: Unitarity Implies No W-III Modes

The one-loop effective action $\Gamma_{1\text{-loop}} = \frac{i}{2} \text{Tr} \ln \hat{H}_{\text{MS}}$ is real (unitarity of the σ -field quantum theory) if and only if all eigenvalues of $\hat{H}_{\text{MS}}$ are non-negative.

Proof. Decompose $\hat{H}_{\text{MS}} = \hat{H}_{\text{MS}}^+ + \hat{H}_{\text{MS}}^-$ into its positive and negative spectral parts (spectral theorem).

$$\text{Tr} \ln \hat{H}_{\text{MS}} = \sum_{\lambda_n > 0} \ln \lambda_n + \sum_{\lambda_n < 0} (\ln |\lambda_n| + i\pi), \quad (33)$$

so $\text{Im} \Gamma_{1\text{-loop}} = -\frac{\pi}{2} \cdot N^-$, where N^- is the number of negative eigenvalues.

By ?? 5.2(ii), W-III modes ($\nu^2 > 1/2$) allow $\langle v | \hat{H}_{\text{MS}} | v \rangle < 0$ for some v , implying $N^- \geq 1$ and hence $\text{Im} \Gamma_{1\text{-loop}} \neq 0$.

The Core Inequality (CI) requires $\hat{H}_{\text{MS}} \geq 0$, i.e., $N^- = 0$, i.e., unitarity. Contradiction. $\square$

Remark 7.1. The non-zero imaginary part of $\Gamma_{1\text{-loop}}$ in the W-III regime corresponds to spontaneous pair creation of σ -particles at rate $\Gamma_{\text{pairs}} = |\text{Im} \Gamma_{1\text{-loop}}|/\mathcal{T}$. The Core Inequality suppresses this rate to zero, which is the cosmological formulation of *informational unitarity*: the universe cannot spontaneously create information from vacuum fluctuations of the σ -field.

8 Route C: Essential Self-Adjointness

Proposition 8.1: Fisher Information Implies Self-Adjointness

If the Core Inequality $\mathcal{I}_{\text{TIC}} \geq 0$ holds strictly (Fisher information is positive), then $\hat{H}_{\text{MS}}$ is essentially self-adjoint on $C_c^\infty(-\infty, 0)$.

Proof. By the Weyl limit-point/limit-circle criterion [1], the operator $\hat{H}_{\text{MS}} = -d^2/d\eta^2 + V_{\text{MS}}(\eta)$ with $V_{\text{MS}}(\eta) = -(\nu^2 - 1/4)/\eta^2$ is essentially self-adjoint on $C_c^\infty(I)$ for $I = (-\infty, 0)$ if and only if the potential is in the *limit-point* case at both endpoints.

At the endpoint $\eta \rightarrow -\infty$: $V_{\text{MS}} \rightarrow 0$, which is in the limit-point case by the Sears criterion (the endpoint is metrically complete and the potential is L_{loc}^2).

At the endpoint $\eta \rightarrow 0^-$: The potential $V_{\text{MS}}(\eta) = -(\nu^2 - 1/4)/\eta^2$ is singular. The Sears–Weyl criterion gives:

Limit-point case (self-adjoint) if and only if $-(\nu^2 - 1/4) \leq 3/4$, i.e., $\nu^2 \leq 1$.

Limit-circle case (failure of self-adjointness) if and only if $-(\nu^2 - 1/4) > 3/4$, i.e., $\nu^2 > 1$.

Now, the Core Inequality (via ?? 5.2) requires $\nu^2 \leq 1/2 < 1$, placing $\hat{H}_{\text{MS}}$ firmly in the limit-point case. Hence $\hat{H}_{\text{MS}}$ is essentially self-adjoint.

Conversely, for $\nu^2 > 1$ (extreme W-III), $\hat{H}_{\text{MS}}$ is in the limit-circle case and is *not* essentially self-adjoint—its deficiency indices are $(1, 1)$ —meaning the quantum theory is ill-defined without additional boundary data. The Core Inequality excludes this regime. $\square$ ■

Remark 8.1. The limit-circle failure at $\eta = 0^-$ for $\nu^2 > 1$ has a profound physical interpretation: the σ -field “falls to the center” (the origin of conformal time, corresponding to the infinite future $t \rightarrow +\infty$ of cosmic time). This is the de Sitter incarnation of the quantum mechanical “fall to the center” for the attractive inverse-square potential. The Core Inequality is precisely the condition that prevents this catastrophe.

9 Connection to the Riemann Hypothesis

9.1 The Spectral Correspondence Hypothesis

We state the following hypothesis, whose proof is deferred to the companion paper (Paper 8 of this series):

Spectral Correspondence Hypothesis (SCH)

$$\text{spec}(\hat{H}_{\text{MS}}) \Big|_{\text{de Sitter fundamental domain}} = \left\{ H^2 \left(\tau_n^2 + \frac{1}{4} \right) \mid \zeta\left(\frac{1}{2} + i\tau_n\right) = 0, \tau_n > 0 \right\}. \quad (\text{SCH})$$

Hypothesis (Spectral Correspondence)

The eigenvalues $\{E_n\}$ of the MS Hamiltonian $\hat{H}_{\text{MS}}$ on the de Sitter fundamental domain $\eta \in (-\pi/H, 0)$ with Dirichlet boundary conditions coincide with the set $\{H^2(\tau_n^2 + 1/4)\}$, where $1/2 + i\tau_n$ ranges over the non-trivial zeros of $\zeta(s)$ on the critical line.

9.2 Proof of Corollary 1.1

Proof of Corollary 1.1. Assume the Spectral Correspondence Hypothesis (SCH).

($\Rightarrow$) Core Inequality $\Rightarrow$ RH.

By ?? 1.1, the Core Inequality requires $\hat{H}_{\text{MS}} \geq 0$, i.e., all eigenvalues $E_n \geq 0$. Under (SCH), $E_n = H^2(\tau_n^2 + 1/4) \geq 0$ holds automatically for $\tau_n \in \mathbb{R}$. But the zeros $1/2 + i\tau_n$ are real- τ (on the critical line) *only if* $\text{Re}(s_n) = 1/2$.

For a zero off the critical line, $s_0 = \sigma_0 + i\tau_0$ with $\sigma_0 \neq 1/2$, the spectral correspondence gives an eigenvalue

$$E_0 = H^2 \left(\tau_0^2 + \sigma_0^2 - \frac{1}{4} + i \cdot 2\tau_0 \left(\sigma_0 - \frac{1}{2} \right) \right). \quad (34)$$

This eigenvalue is *complex*. But $\hat{H}_{\text{MS}}$ is self-adjoint (by Proposition 8.1), so all eigenvalues are real. Contradiction. Hence $\sigma_0 = 1/2$ for all non-trivial zeros.

($\Leftarrow$) **RH $\Rightarrow$ Core Inequality.**

If all non-trivial zeros satisfy $\text{Re}(s_n) = 1/2$ (RH holds), then all eigenvalues $E_n = H^2(\tau_n^2 + 1/4) > 0$ (since $\tau_n \in \mathbb{R}^+$). Hence $\hat{H}_{\text{MS}} > 0$ (strictly positive), which implies $\mathcal{Q}[v] > 0$ for all $v \neq 0$, giving the Core Inequality.

This establishes the equivalence:

Core Inequality (TIC/CIT) $\iff$ self-adjointness of $\hat{H}_{\text{MS}}$ $\iff$ Riemann Hypothesis
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(35)

under the Spectral Correspondence Hypothesis. $\square$ ■

9.3 Schematic Diagram of Logical Dependencies

$$\begin{array}{ccccc}
\text{Axioms 1-4 (TIC/CIT)} & \Rightarrow & \text{Core Inequality} & \Rightarrow & \hat{H}_{\text{MS}} \geq 0 \\
& & \Updownarrow & & \Updownarrow \\
\text{RH} & \stackrel{(\text{SCH})}{\iff} & \text{no W-III modes} & \Leftrightarrow & \mathcal{Q}[v] \geq 0 \ \forall v \\
& & \Updownarrow & & \Updownarrow \\
& & \text{Unitarity} & \Leftrightarrow & \text{Im } \Gamma_{1\text{-loop}} = 0
\end{array}$$

10 Sharpness and Optimality

Theorem 10.1: Optimality of the Hardy Bound

The threshold $\nu_{\text{cr}}^2 = 1/2$ (i.e., $\omega_\sigma^{*2} = 7H^2/4$) in ?? 1.1 is optimal: for any $\delta > 0$, there exists a TIC/CIT σ -configuration with $\nu^2 = 1/2 - \delta$ (W-I, just above threshold) that saturates the Hardy inequality arbitrarily closely.

Proof. For $\nu^2 = 1/2 - \delta$, the quadratic form is $\mathcal{Q}[v] \geq (1/2 - \nu^2) \int v^2/\eta^2 d\eta = \delta \int v^2/\eta^2 d\eta \geq 0$. For the sequence of test functions $v_R = R^{-\varepsilon} |\eta|^{1/2+\varepsilon} \chi_{1/R,R}$ (with $\varepsilon \rightarrow 0$ and $R \rightarrow \infty$ simultaneously), $\mathcal{Q}[v_R]/\|v_R\|^2 \rightarrow \delta$ from above, showing the bound is saturated in the limit. The minimum is not achieved in L^2 (cf. sharpness of Hardy, ?? 5.1). $\square$ ■

11 Summary and Discussion

We have given a complete, self-contained mathematical proof that W-III modes of the TIC/CIT σ -field in the FLRW background are strictly prohibited by the Core Inequality. The proof rests on three independent pillars:

1. **Route A (Hardy inequality):** The Core Inequality is equivalent to positive semi-definiteness of the Mukhanov–Sasaki Hamiltonian $\hat{H}_{\text{MS}}$. By the Hardy inequality on the half-line, this holds if and only if $\nu^2 \leq 1/2$, i.e., $\omega_\sigma^2 \geq 7H^2/4$. W-III modes violate this bound, and the violation is demonstrated by an explicit test function (20) for which $\langle v | \hat{H}_{\text{MS}} | v \rangle < 0$.
2. **Route B (Unitarity):** W-III modes generate a non-zero imaginary part in the one-loop effective action, corresponding to spontaneous pair creation. This violates the unitarity of the TIC/CIT quantum theory, which follows from the Core Inequality.
3. **Route C (Self-adjointness):** The Core Inequality (positive Fisher information) implies that $\hat{H}_{\text{MS}}$ is essentially self-adjoint. W-III modes with $\nu^2 > 1$ place $\hat{H}_{\text{MS}}$ in the limit-circle case, destroying self-adjointness and making the quantum theory ill-defined.

Under the Spectral Correspondence Hypothesis, all three routes converge on the equivalence (35): the Core Inequality is equivalent to the Riemann Hypothesis. The proof of the Spectral Correspondence itself will be the subject of Paper 8 of this series.

Open Questions

1. Can the threshold $\omega_\sigma^{*2} = 7H^2/4$ be tightened to $9H^2/4$ (the de Sitter BF bound) using the TIC/CIT axioms alone, without appeal to the Core Inequality?
2. Does the prohibition of W-III modes extend to non-Gaussian (interacting) σ configurations?
3. What is the precise nature of the Spectral Correspondence between $\hat{H}_{\text{MS}}$ and the Riemann zeros?

Acknowledgements

The author thanks the TIC/CIT review consortium for critical commentary on Papers 1–6, and acknowledges the foundational contributions of B. Simon, M. Reed, and T. Kato to the mathematical methods employed here.

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A Verification of the Test Function Estimate

We verify that the estimate eq. (24) is correct by explicit calculation for the smooth cutoff version of v_ε .

Let $\chi \in C_c^\infty(1, 2)$ with $0 \leq \chi \leq 1$, $\chi \equiv 1$ on $[5/4, 7/4]$. For $R > 1$ and $\delta \in (0, 1/4)$, define

$$v_\varepsilon(\eta) = |\eta|^{1/2+\varepsilon} \cdot \chi\left(\frac{|\eta| - \delta}{\delta}\right) \cdot \chi\left(\frac{R - |\eta|}{R/4}\right).$$

By direct computation:

$$\begin{aligned} \int_{-\infty}^0 (v'_\varepsilon)^2 d\eta &= (1/2 + \varepsilon)^2 \int_{\delta}^R |\eta|^{2\varepsilon-1} d\eta + O(\delta^{2\varepsilon} + R^{-(1-2\varepsilon)}) \\ &= \frac{(1/2 + \varepsilon)^2}{2\varepsilon} [1 + O(\delta^{2\varepsilon})], \\ \int_{-\infty}^0 \frac{v_\varepsilon^2}{\eta^2} d\eta &= \int_{\delta}^R |\eta|^{2\varepsilon-1} d\eta + O(\delta^{2\varepsilon}) = \frac{1}{2\varepsilon} [1 + O(\delta^{2\varepsilon})]. \end{aligned}$$

Therefore:

$$\mathcal{Q}[v_\varepsilon] = \frac{1}{2\varepsilon} [(1/2 + \varepsilon)^2 - (\nu^2 - 1/4)] + O(\delta^{2\varepsilon}) = \frac{\frac{1}{2} - \nu^2 + \varepsilon + \varepsilon^2}{2\varepsilon} + O(\delta^{2\varepsilon}).$$

For $\nu^2 > 1/2$ and $\varepsilon = (\nu^2 - 1/2)/2 < 1$:

$$\mathcal{Q}[v_\varepsilon] = \frac{-(\nu^2 - 1/2)/2 + (\nu^2 - 1/2)^2/4}{2 \cdot (\nu^2 - 1/2)/2} + O(\delta^{\nu^2 - 1/2}) = \frac{(\nu^2 - 1/2)/4 - 1/2}{2} + O(\delta^{\nu^2 - 1/2}).$$

Taking $\delta \rightarrow 0$ with $\nu^2 > 1/2$ fixed:

$$\mathcal{Q}[v_\varepsilon] \rightarrow \frac{(\nu^2 - 1/2)/4 - 1/2}{2} = \frac{\nu^2 - 5/2}{8}.$$

For $1/2 < \nu^2 < 5/2$ (which includes all physically relevant cases): $\mathcal{Q}[v_\varepsilon] < 0$. $\square$

B The Inverse-Square Potential: Classical Results

The operator $L_c = -d^2/dx^2 - c/x^2$ on $(0, \infty)$ is the *Calogero–Marchioro operator*. The following are classical:

- For $c \leq 1/4$: L_c is essentially self-adjoint on $C_c^\infty(0, \infty)$ and $L_c \geq 0$ (Hardy bound).
- For $c > 1/4$: L_c is in the limit-circle case at $x = 0$ and requires a self-adjoint extension. For $1/4 < c < 3/4$, the Friedrichs extension is positive; for $c \geq 3/4$, some extensions may have negative spectrum.
- The threshold $c = 1/4$ corresponds in our notation to $\nu^2 - 1/4 = 1/4$, i.e., $\nu^2 = 1/2$.

These results confirm that the threshold $\nu_{\text{cr}}^2 = 1/2$ in ?? 1.1 is the exact mathematical boundary between the stable (Core Inequality satisfied) and unstable (Core Inequality violated) regimes of the TIC/CIT σ -field.